

IMO Training – Number Theory

Order, Valuation and Diophantine Equations

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Notes

Unless otherwise specified, let p be a prime number.

Theorem. (Fermat's Little Theorem) $\forall a \in \mathbb{Z}, a^p \equiv a \pmod{p}$. Equivalently, $\forall a \in \mathbb{Z}$ with $(a, p) = 1$, $a^{p-1} \equiv 1 \pmod{p}$.

Definition. (Euler Totient Function) $\forall n \in \mathbb{N}$, $\varphi(n) = \#\{m \in \mathbb{N} \mid 1 \leq m < n, (m, n) = 1\}$. Equivalently, for $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, where $p_1, p_2, \dots, p_k$ being distinct primes, and integers $a_1, a_2, \dots, a_k \geq 1$, $\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$.

Theorem. (Fermat-Euler Theorem) $\forall a \in \mathbb{Z}, n \in \mathbb{N}$ s.t. $(a, n) = 1$, $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Theorem. (Wilson's Theorem) p is a prime iff $(p-1)! \equiv -1 \pmod{p}$.

Definition. (Order and Primitive Roots) For $a \in \mathbb{Z}, n \in \mathbb{N}$ with $(a, n) = 1$, the *order of a modulo n* ($\text{ord}_n(a)$) is defined by the minimal $m \in \mathbb{N}$ s.t. $a^m \equiv 1 \pmod{n}$. Furthermore, if $\text{ord}_n(a) = \varphi(n)$, then a is a *primitive root modulo n* .

Definition. (Valuation) For $m \in \mathbb{N}$, the *m -adic valuation* of n is the $a \in \mathbb{N}$ s.t. $m^a \mid n$ and $m^{a+1} \nmid n$.

Example. Find $\text{ord}_3(2)$.

Example. Find $\text{ord}_{17}(3)$.

Example. Find $\text{ord}_{101}(2)$.

Example. Find all primitive roots of 29 between 1 and 28.

Example. Prove that if $a, m, n \in \mathbb{N}$, where a, m are coprime, then $a^n \equiv 1 \pmod{m}$ iff $\text{ord}_m(a) \mid n$.

Example. Prove that for any prime p , every prime divisor of $2^p - 1$ is greater than p .

Example. Let p be an odd prime, and let q, r be primes s.t. $p \mid q^r + 1$. Prove that $2r \mid p - 1$ or $p \mid q^2 - 1$.

Example. Let $a, n \in \mathbb{N}$ with $a > 1$. If p is an odd prime s.t. $p \mid a^{2^n} + 1$, prove that $2^{n+1} \mid p - 1$.

Example. (Putnam 1972) Let $n \in \mathbb{Z}, n \geq 2$. Prove that $n \nmid 2^n - 1$.

Example. Prove that for all $a, n \in \mathbb{N}$ with $a > 1, n \mid \phi(a^n - 1)$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $3x + 4y = 1$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $11x + 22y = 13$.

Example. Find all $n \in \mathbb{N}$ s.t. $2^8 + 2^{11} + 2^n$ is a perfect square.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $x^3 + y^3 = 2006$.

Example. Find all $x, y \in \mathbb{Z}$ s.t. $y^2 = x^3 + 7$.

Example. Find all $x, y \in \mathbb{Z}_{\geq 0}$ s.t. $1 + 3^x = 2^y$.

Example. Find all $x, y \in \mathbb{Z}_{\geq 0}$ s.t. $1 + 2^x = 3^y$.

Example. Find all $x, y, z \in \mathbb{N}$ s.t. $3^x + 4^y = 5^z$.

Exercises

1. (AIME 2001) How many positive integer multiples of 1001 can be expressed in the form $10^j - 10^i$, where i and j are integers and $0 \leq i < j \leq 99$?
2. (IMO 1978) Let m and n be natural numbers such that $1 \leq m < n$. In their decimal representations, the last three digits of 1978^m are equal, respectively, to the last three digits of 1978^n . Find m and n such that $m + n$ has its least value.
3. (IMO 1990) Find all positive integers $n > 1$ such that $\frac{2^n + 1}{n^2}$ is an integer.
4. (Bulgaria 1996) Find all primes p, q s.t. $pq \mid (5^p - 2^p)(5^q - 2^q)$.
5. (USA TST 2003) Find all ordered prime triples (p, q, r) s.t. $p \mid q^r + 1$, $q \mid r^p + 1$, $r \mid p^q + 1$.
6. (China 2009) Find all primes p, q s.t. $pq \mid 5^p + 5^q$.
7. (China 2005) Find all $x, y, z, w \in \mathbb{Z}_{\geq 0}$ s.t. $2^x 3^y - 5^z 7^w = 1$.
8. (Bulgaria 2006) Find all $t, x, y, z \in \mathbb{N}$ s.t. $2^t = 3^x 5^y + 7^z$.

Extra Space